

Machine Learning Assignment

Customer Churn Prediction & Revenue Forecasting using Supervised Learning

Overview

You are an ML Engineer at **TeleConnect**, a telecom company with 50,000+ customers. The company is losing customers and struggling with revenue forecasting. Apply your supervised learning knowledge to build a complete ML pipeline that solves two real-world business problems:

Problem 1 (Classification): Predict which customers will churn

Problem 2 (Regression): Forecast monthly revenue per customer

You must implement **all major supervised learning algorithms**, compare their performance, and provide actionable business recommendations.

Dataset

Use the **Telco Customer Churn** dataset from Kaggle:

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

The dataset contains **7,043 customer records** with 21 features:

Feature	Type	Description
customerID	string	Unique customer identifier
gender	categorical	Male / Female
SeniorCitizen	binary	0 or 1
Partner	categorical	Yes / No
Dependents	categorical	Yes / No
tenure	numeric	Months with the company
PhoneService	categorical	Yes / No
MultipleLines	categorical	Yes / No / No phone service

InternetService	categorical	DSL / Fiber optic / No
OnlineSecurity	categorical	Yes / No / No internet service
OnlineBackup	categorical	Yes / No / No internet service
DeviceProtection	categorical	Yes / No / No internet service
TechSupport	categorical	Yes / No / No internet service
StreamingTV	categorical	Yes / No / No internet service
StreamingMovies	categorical	Yes / No / No internet service
Contract	categorical	Month-to-month / One year / Two year
PaperlessBilling	categorical	Yes / No
PaymentMethod	categorical	Electronic check / Mailed check / Bank transfer / Credit card
MonthlyCharges	numeric	Monthly charge amount
TotalCharges	numeric	Total charges to date
Churn	binary	Yes / No TARGET - Classification

Note: MonthlyCharges is the **TARGET for regression**. When building the regression model, exclude it from features and use remaining columns to predict it.

Task 1: Exploratory Data Analysis (EDA)

15 Marks

- Handle missing values with justification for chosen strategy
- Detect and treat outliers (use IQR or Z-score, justify your choice)
- Analyze class imbalance in the Churn column (show distribution plot)
- Correlation analysis between numerical features (heatmap)
- Feature distribution analysis (histograms, box plots)
- Bivariate analysis: how each feature relates to Churn (at least 5 features)
- Document at least **3 business insights** from EDA

Task 2: Feature Engineering & Preprocessing

15 Marks

- Encode categorical variables (use **both** Label Encoding and One-Hot Encoding, explain when to use which)
 - Feature scaling (apply **both** StandardScaler and MinMaxScaler, compare results)
 - Create at least **3 new derived features**, for example:
 - $\text{AvgMonthlySpend} = \text{TotalCharges} / \text{tenure}$
 - $\text{ServiceCount} = \text{count of subscribed services}$
 - $\text{ContractValue} = \text{MonthlyCharges} * \text{remaining_contract_months}$
 - Feature selection using at least **2 methods**:
 - Correlation-based filtering
 - Recursive Feature Elimination (RFE)
 - Feature importance from tree-based models
 - Mutual Information
 - Handle class imbalance using at least **2 techniques**:
 - SMOTE
 - Random Undersampling
 - Class weights
 - Split data: **70% train / 15% validation / 15% test** (use stratified split for classification)
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Task 3: Classification — Churn Prediction

30 Marks

Build and compare **all 5** of the following classifiers:

#	Algorithm	Required
1	Logistic Regression	Required
2	Decision Tree Classifier	Required
3	Random Forest Classifier	Required
4	Support Vector Machine (SVM)	Required
5	K-Nearest Neighbors (KNN)	Required

For each algorithm you must:

- Train on training set, tune on validation set
- Perform hyperparameter tuning using **GridSearchCV** or **RandomizedSearchCV** (show parameter grid)
- Report these metrics on the **test set**:
 - Accuracy
 - Precision, Recall, F1-Score (per class)
 - ROC-AUC Score
 - Confusion Matrix (as heatmap)
 - ROC Curve (all models on same plot)
- Explain **why** each algorithm performed the way it did (not just numbers, explain the reasoning)

Final Comparison Table (fill in your results):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Training Time
Logistic Regression						
Decision Tree						
Random Forest						
SVM						
KNN						

- Select the **best model** with detailed justification
 - For the best model, show **feature importance** (top 10 features with bar chart)
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Task 4: Regression — Revenue Forecasting

25 Marks

Predict `MonthlyCharges` using the remaining features. Build and compare **all 7** of the following:

#	Algorithm	Required
1	Linear Regression	Required
2	Ridge Regression (L2)	Required
3	Lasso Regression (L1)	Required
4	ElasticNet	Required
5	Decision Tree Regressor	Required
6	Random Forest Regressor	Required
7	Support Vector Regressor (SVR)	Required

For each algorithm you must:

- Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV
- Report these metrics on the **test set**:
 - MAE (Mean Absolute Error)
 - MSE (Mean Squared Error)
 - RMSE (Root Mean Squared Error)
 - R-squared (R^2)
 - Adjusted R-squared
- Plot **Actual vs Predicted** scatter plot for each model
- Plot **Residual distribution** for the best model

Final Comparison Table (fill in your results):

Model	MAE	MSE	RMSE	R^2	Adj. R^2	Training Time
Linear Regression						
Ridge						
Lasso						
ElasticNet						

Decision Tree
Random Forest
SVR

- Select the **best model** with detailed justification
 - Analyze residuals: are they normally distributed? Any patterns?
 - Show **coefficient analysis** for linear models (which features have highest impact?)
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Task 5: Model Interpretation & Business Recommendations

15 Marks

- Use **SHAP values** to explain predictions of the best classification model:
 - Global feature importance (SHAP summary plot)
 - Local explanation for at least 2 individual predictions (one churned, one retained)
- Use **Partial Dependence Plots (PDP)** for top 3 features
- Answer these business questions with data evidence:
 1. What are the top 5 factors driving customer churn?
 2. Which customer segments are at highest risk?
 3. What pricing strategy would you recommend based on the regression model?
 4. If the company can only intervene with 100 customers, which ones should they target and why?
- Provide a **cost-benefit analysis**: assuming it costs \$50 to retain a customer and a churned customer costs \$500 in lost revenue, what is the ROI of using your model vs. random targeting?

Submission Instructions

GitHub Repository Structure

```
your-github-username/teleconnect-ml-assignment/
|
|-- README.md                # Project overview, setup, results
summary
|-- requirements.txt         # All Python dependencies with
versions
|-- .gitignore              # Ignore data/, *.pyc,
.ipynb_checkpoints
|
|-- data/
|   |-- raw/                # Original dataset
|   |-- processed/          # Cleaned & feature-engineered data
|   |-- README.md           # Data dictionary and source link
|
|-- notebooks/
|   |-- 01_EDA.ipynb        # Exploratory Data Analysis
```

```
| |-- 02_preprocessing.ipynb # Feature Engineering &
Preprocessing
| |-- 03_classification.ipynb # All classification models
| |-- 04_regression.ipynb # All regression models
| |-- 05_interpretation.ipynb # SHAP, PDP, business
recommendations
|
|-- src/
| |-- __init__.py
| |-- data_loader.py # Functions to load and validate
data
| |-- preprocessing.py # Encoding, scaling, feature
engineering
| |-- classifiers.py # Classification model training &
eval
| |-- regressors.py # Regression model training & eval
| |-- evaluation.py # Metrics, plots, comparison tables
| |-- utils.py # Helper functions
|
|-- models/
| |-- best_classifier.pkl # Saved best classification model
| |-- best_regressor.pkl # Saved best regression model
| |-- scaler.pkl # Fitted scaler
| |-- encoder.pkl # Fitted encoder
|
|-- reports/
| |-- figures/ # All plots and charts (PNG/SVG)
| |-- classification_report.md # Detailed classification results
| |-- regression_report.md # Detailed regression results
|
|-- tests/
| |-- test_preprocessing.py # Unit tests for preprocessing
| |-- test_evaluation.py # Unit tests for evaluation metrics
```

Repository README Must Include

1. **Project Title & Description** (2-3 lines)
2. **Dataset** — source link and brief description
3. **Installation & Setup**

```
git clone https://github.com/your-username/teleconnect-ml-assignment.
cd teleconnect-ml-assignment
python -m venv venv
```

```
source venv/bin/activate # or venv\Scripts\activate on Windows
pip install -r requirements.txt
```

4. **How to Run** — step-by-step commands to reproduce results
5. **Results Summary** — best model, key metrics, one key insight
6. **Project Architecture** — folder structure explanation
7. **Tech Stack** — libraries used and why

Git Practices Required

- Minimum **10 meaningful commits** (not bulk commits)
- Use clear commit messages:

```
feat: add EDA notebook with correlation analysis
feat: implement SMOTE for class imbalance handling
fix: correct one-hot encoding for multi-category features
docs: add results summary to README
refactor: extract preprocessing into reusable functions
```

- Use **branches** for each major task:

```
git checkout -b feature/eda
git checkout -b feature/preprocessing
git checkout -b feature/classification
git checkout -b feature/regression
git checkout -b feature/interpretation
```

- Merge each branch into `main` via pull request (even if solo)

Code Quality Requirements

- Follow **PEP 8** coding standards
- Add **docstrings** to all functions
- No hardcoded file paths (use `os.path` or `pathlib`)
- Reproducibility: set `random_state=42` everywhere
- All notebooks must run **top-to-bottom** without errors
- Use `requirements.txt` with pinned versions:

```
pandas==2.1.4
numpy==1.26.2
scikit-learn==1.3.2
matplotlib==3.8.2
seaborn==0.13.0
shap==0.44.0
jupyter==1.0.0
```


Evaluation Criteria

Criteria	Marks	What We Look For
EDA	15	Depth of analysis, meaningful visualizations, business insights
Feature Engineering	15	Creativity, justification, proper preprocessing pipeline
Classification	30	All 5 models implemented, proper tuning, correct evaluation, clear comparison
Regression	25	All 7 models implemented, proper tuning, residual analysis
Interpretation	15	SHAP usage, business recommendations backed by data
Bonus	+10	Unit tests, clean modular code in src/, CI pipeline, Docker

Total: 100 Marks (+10 Bonus)

Important Notes

Plagiarism will result in immediate disqualification. We check for code similarity across submissions and against public repositories.

- You may use documentation and learning resources, but the implementation must be your own.
- If you use any code snippet from external sources, cite it in comments.
- Focus on **understanding and explanation** over just getting high accuracy.
- We value clean, well-documented code over complex solutions.
- Do **NOT** use AutoML libraries (auto-sklearn, TPOT, etc.) — implement manually.
- All plots must be visible in the notebooks (use `%matplotlib inline`).

Submission Format

- Push your complete code to a **public GitHub repository**
- Share the repository link before the deadline
- Ensure all notebooks execute without errors

— Good Luck! —